

Large Multimodal Model-Driven Clinical Decision Support Systems: Mapping Architectures to Decision Requirements—A Scoping Review Protocol

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1. Introduction

1.1. Background

The current landscape of Large Multimodal Model-based clinical decision support systems (LMM-CDSS) is characterized by a tension between unprecedented technological acceleration and pronounced systemic fragmentation. As generative AI architectures move from research proofs-of-concept to real-world clinical deployment at an increasingly rapid rate [1, 2], this progress unfolds within a growing, interdisciplinary void. The required convergence of computer science, clinical medicine, ethics, and regulatory science [3, 4] has produced substantial heterogeneity in both data modalities, [spanning unstructured EHR data to complex multi-sensor time series](#), and methodological approaches, including Retrieval-Augmented Generation

(RAG) and multi-agent paradigms [1, 5]. Although this diversity is a powerful driver of innovation, it has advanced faster than the formation of shared conceptual frameworks, leaving a significant gap in our understanding of how these disparate components should be integrated into coherent decision-making archetypes that support safe clinical use.

To verify the originality of this study and ensure a non-redundant contribution to the field, a multi-stage preliminary research strategy was executed in February 2026. The preliminary inquiry began with a systematic search for published scoping and systematic reviews at the intersection of clinical CDSS and LMMs. The search scope was strictly constrained to clinical decision-making, deliberately excluding administrative automation, medical training strategies, and patient-facing AI applications. A specific emphasis was placed on the governing attributes of clinical decisions, examining the influence of medical specialties and the inherent complexity of decision-making tasks. The inquiry utilized PubMed (biomedical), IEEE Xplore (technological), and Web of Science (multidisciplinary) databases, supplemented by a targeted audit of the *JBIR Evidence Synthesis* journal and formal registry searches across OSF Registries and PROSPERO. To ensure a comprehensive identification of both protocols and published work, this traditional search was augmented by AI-driven discovery using Perplexity and NotebookLM. These tools were employed to identify additional references that may have escaped standard database indexing, including emerging technical reports and recently registered protocols. These collective verification steps confirmed that no existing or ongoing reviews address the specific aim of this study, namely the application of a decision-centric framework aligned with LMM

architectural functionalities.

Preliminary research has established that the landscape of Large Multimodal Models (LMMs) in healthcare is characterized by significant technical velocity. Foundational scoping and systematic reviews have successfully mapped the global distribution of AI applications across broad medical specialties, identifying a high degree of performance heterogeneity across diverse clinical tasks [6, 7]. Recent benchmarks have introduced practical three-tier frameworks to align agent architectures with task complexity, ranging from discrete micro-tasks to multi-agent ecosystems [8]. While these studies provide essential data on architectural alignment and the role of customization pipelines like Retrieval-Augmented Generation (RAG) [9], they typically treat medical specialties as static evaluation categories rather than active governors of decision attributes.

Our protocol proposes to transcend broad complexity tiers by employing a decision-centric framework that analyzes how the specific nature of a medical specialty dictates decision attributes, and therefore architectural requirements. Current literature distinguishes between single-agent systems for narrow, vertical domains and multi-agent ecosystems for high-stakes, horizontal problems requiring the synthesis of conflicting evidence [8, 10]. However, a critical gap remains in formalizing how these decisions demand vertical pathophysiological reasoning versus horizontal multidisciplinary integration. While handling long temporal sequences and temporal reasoning are recognized as significant future directions [11], existing reviews often treat clinical tasks as one-off judgments. The current protocol addresses this limitation by distinguishing between static clinical snapshots and longitudinal reasoning

trajectories over evolving patient histories.

To maintain high precision and clinical relevance, this review is strictly delimited to the role of the Digital Practitioner in direct clinical decision-making. This focus distinguishes the current scope from broader foundational works that include healthcare administrative tasks (e.g., billing, scheduling), medical education and licensing exams, or patient-facing self-triage tools [6, 12, 13]. Finally, whereas contemporary research evaluates specialized fine-tuned models primarily through the lens of training efficiency [9], our approach analyzes how specific clinical case types and specialty-driven constraints force architectural evolution. By centering on these functional decision archetypes, the review provides a more fine-grained taxonomy of procedural complexity than is currently available in the literature.

1.2. Objectives

The primary objective of this scoping review is to map how clinical decision characteristics act as the governor of system design in LMM-driven clinical decision support systems.

Specifically, the review aims to:

- **Characterize the Decision space:** Identify and formally represent the principal factors that collectively specify the fundamental properties of a clinical decision space. These include, for example, the level and type of inference required in highly specialized clinical contexts as compared to the degree of integration necessary in cross-specialty or multi-disciplinary systems; the applicable regulatory and normative constraints (e.g., EU AI Act, specialty-specific clinical guidelines); the

level of clinical criticality and risk; as well as additional relevant determinants that shape the operational, ethical, and legal boundaries of decision-making processes.

- **Develop a Taxonomy of the Clinical Decision Space:** Construct a taxonomy of decision space archetypes that incorporates the identified characterization. This entails analyzing how the interaction between these factors collectively determines the structural specifications and constraints of clinical decision-making tasks.
- **Analyze the Architectural Alignment to Clinical Constraints:** Identify how specific clinical decision factors force technical evolutions in system design by mapping the problem/decision Space and the Solution Space. This involves investigating how high-complexity clinical requirements necessitate the integration of advanced architectural paradigms, such as Multi-Agent Systems (MAS) for integrative breadth, temporal modules for longitudinal reasoning, and Advanced Explainability (XAI) or Retrieval-Augmented Generation (RAG) for fact-checking and clinical justification.
- **Evaluate the Determinants of Real-World Clinical Deployment:** Examine how the alignment between decision characteristics and system architecture influences the transition from experimental prototypes to clinical-grade implementation. This involves analyzing how factors such as Advanced Explainability (XAI), fact-checking mechanisms such as RAG, and the management of high-stakes clinical actions drive the development of deployment-ready solutions capable

of maintaining logical fidelity and trustworthiness across heterogeneous medical workflows.

1.3. Review Questions

The following research questions will guide this scoping review:

Primary Question:

- How do clinical decision characteristics act as the governor for the design and implementation of Large Multimodal Model (LMM)-driven clinical decision support systems?

Sub-questions:

1. **Characterizing the Decision Space:** What are the principal factors (e.g., specialized inference vs. multi-disciplinary integration, regulatory constraints, and clinical risk) that define the operational, ethical, and legal boundaries of a clinical decision space?
2. **Taxonomy of the Decision Space:** How can these identified factors be used to construct a taxonomy of clinical decision archetypes that reflects the structural specifications and constraints of different medical tasks?
3. **Architectural Alignment:** How do specific clinical decision factors force technical evolutions in system design, specifically requiring advanced paradigms such as Multi-Agent Systems, temporal modules, or fact-checking mechanisms like RAG?
4. **Deployment Determinants:** How does the alignment between decision characteristics and system architecture influence the transition

from experimental prototypes to clinical-grade implementations that maintain trust and logical fidelity?

2. Inclusion Criteria

The inclusion criteria for this scoping review are structured according to the PCC (Population, Concept, Context) framework to ensure direct alignment with the study’s decision-centric objectives.

2.1. Population

This review considers studies involving Digital Practitioners, defined as licensed healthcare professionals (physicians, specialists, nurses, or residents) and multidisciplinary clinical teams utilizing LMM-CDSS within a professional workflow.

- **Inclusion:** Clinicians across all medical specialties and cross-specialty diagnostic teams.
- **Exclusion:** Studies focused on patient-facing applications (self-diagnosis, symptom checkers), on undergraduate medical students (educational tools), or on administrative staff (billing/scheduling).

2.2. Concept

The core concept is the Large Multimodal Model-driven Clinical Decision Support System (LMM-CDSS). The category encompasses Large Multimodal Models (LMMs) and Large Language Models (LLMs) used to process clinical data (text, images, vitals, etc.) for clinical decision support.

- **Inclusion:** The system must produce patient-specific recommendations intended to support a clinical decision (e.g., diagnosis, treatment, prognosis, triage, or discharge planning).
- **Exclusion:**
 - *Non-generative AI:* standalone traditional machine learning (e.g., Random Forest, CNN-only, SVM) or rule-based expert systems lacking a generative transformer-based core.
 - *Operational and administrative tasks:* tools used exclusively for administrative automation (e.g., automated scribing, billing, ICD coding) or general medical education and information retrieval without a patient-specific recommendation or decision context.

Terminological note on LMM vs. LLM: This review adopts *Large Multimodal Model* (LMM) as its primary term, following the World Health Organization’s *Guidance on Large Multi-Modal Models* [14], which reflects the architectural shift beyond text-only processing toward native multimodal integration. Because the indexed literature predominantly uses *Large Language Model* (LLM) even for multimodal systems, both terms are retained in all search strings. For inclusion purposes, LLM and LMM designate the same class of generative, transformer-based architectures; the distinction is terminological, not exclusionary.

2.3. Context

The review encompasses all clinical environments where medical decisions are formulated, ranging from traditional infrastructure (Hospitals, Clinics,

Emergency Departments) to resource-constrained settings (Rural, Remote, or Mobile-based care). The setting also includes computational environments to accommodate benchmark and validation studies that utilize standardized medical datasets.

- **Inclusion:** This scoping review adopts a global perspective and spans all medical specialties. To trace how generalism and specialization have developed, the review explicitly focuses on two decision frameworks: specialized decisions and cross-specialization decisions.
- **Exclusion:** Non-clinical hospital management (resource allocation, bed management), public health policy modeling without direct practitioner interaction, and medical imaging software that performs only segmentation without a decision-support component.

2.4. Types of Sources and Evidence

This review will consider primary and secondary research published in peer-reviewed journals and conference proceedings.

- **Language:** Only studies published in English will be included.
- **Date Range:** The search will be limited to evidence published from 2023 to the present, reflecting the emergence of multimodal architectural paradigms.
- **Evidence Types:** Peer-reviewed primary research, conference papers (e.g., NeurIPS, MICCAI), and systematic/scoping reviews for reference harvesting.

- **Grey Literature Policy:** Pre-print repositories such as arXiv are excluded from the primary search for two methodological reasons: pre-prints have not undergone peer review, introducing a quality asymmetry with indexed sources; and this review targets systems with sufficient clinical evaluation, work that predominantly appears in peer-reviewed venues. Grey literature is not categorically excluded, however, pre-prints or technical reports identified through citation snowballing or cited within primary sources will be assessed against the inclusion criteria and considered for inclusion.

3. Methodology

3.1. Framework and Design

Given the characteristics of the LMM-CDS research landscape, this study employs a *scoping review* design to comprehensively synthesize the extant literature, with an emphasis on systematic evidence mapping and the identification of knowledge gaps. To balance methodological rigor with feasibility, a formal critical appraisal of individual study quality is not undertaken, consistent with prevailing conventions for scoping reviews that prioritize evidence mapping and gap detection over the evaluation of intervention effectiveness [15].

This scoping review is conducted in accordance with the Joanna Briggs Institute (JBI) methodological framework [15] and reported following the PRISMA-ScR guidelines to ensure comprehensive transparency [16]. To address the integration of artificial intelligence (AI) as a methodological tool, the review incorporates the PRISMA-trAIce extension which is a discipline-

agnostic checklist designed to facilitate the transparent reporting of AI-assisted steps such as discovery, screening, and data handling across all stages of evidence synthesis [17]. A single PRISMA-trAIce reporting commitment governs every AI-assisted step in this protocol: at each AI use, the tool or model name, version, prompting strategy (system and user prompts), and date of use are recorded in the screening database or, for discovery steps, in the final research paper. The collective application of these standards supports methodological reproducibility while providing a foundation for mapping the current evidence base, delineating research gaps, and informing the direction of future research.

This protocol has been registered on the Open Science Framework (DOI: 10.17605/OSF.IO/K7CDY). Subsequent amendments to the protocol will be recorded as new versions in the same OSF registration to preserve a transparent change history.

3.2. Search Strategy

The search strategy will follow the JBI three-step process to ensure a comprehensive, transparent, and reproducible identification of both published and grey literature [15]. In alignment with the Decision-Centric Framework, the strategy is calibrated to capture the intersection of clinical reasoning trajectories and LMM architectural responses.

3.2.1. Initial Exploratory Search

The initial exploratory search, executed in February 2026, served two complementary functions aligned with JBI step one: confirming the absence of overlapping protocols, and harvesting the high-frequency index terms and

keywords that underpin the comprehensive search strings described in Section 3.2.2. This stage functioned as the foundational anchor for defining the boundaries of both the clinical Problem Space and the technical Solution Space. A systematic audit was conducted across PubMed, IEEE Xplore, Web of Science, and the JBI Evidence Synthesis journal. Formal registry searches were simultaneously performed within OSF Registries and PROSPERO to verify that no existing or in-progress protocols address the integration of a decision-centric framework with Large Multimodal Model (LMM) architectural functionalities.

The exploratory search identified several foundational works, particularly regarding Multi-Agent Systems (MAS) in high-criticality healthcare. This preliminary inquiry was instrumental in refining the final search strings through the identification of high-frequency index terms; it also allowed for the precise establishment of inclusion constraints, in particular the exclusion of administrative automation and patient-facing applications, to ensure a dedicated focus on the workflow of the Digital Practitioner.

To reduce the likelihood of missing new technical reports or recently registered protocols that have not yet been indexed in conventional databases, this stage was supplemented with AI-based discovery using Perplexity and NotebookLM. These tools supported a broad mapping of the current state-of-the-art literature, with particular attention to how clinical requirements are presently driving technical shifts toward Retrieval-Augmented Generation (RAG) and specialized inferential architectures. The sources identified by these systems were then examined to determine whether they should be incorporated into the preliminary findings presented in the introduction. [The](#)

product versions, prompting strategies, and exact dates of use for Perplexity and NotebookLM during this exploratory phase will be reported in the final research paper, in line with the PRISMA-trAIce commitment stated in Section 3.1.

3.2.2. Comprehensive Search Strategy

To capture the multidimensional intersection of clinical reasoning and architectural responses, three primary databases were selected based on their specific indexing strengths and the necessity of cross-validating the Problem-Solution overlap. This triad ensures that the search captures high-impact medical research, current technical innovations, and the interdisciplinary synthesis required for a decision-centric review.

PubMed (National Library of Medicine) serves as the primary reference source for the clinical domain. It offers dedicated indexing of peer-reviewed biomedical literature and MEDLINE records, ensuring that the studies identified are firmly aligned with established clinical practice workflows. Its use is essential for specifying the boundaries of the Digital Practitioner role and for articulating the established concept of a Clinical Decision Support System, which is represented as a medical subheading in MeSH.

Search String (PubMed):

```
("Decision Support Systems, Clinical"[MeSH]) AND ("Large  
language model"[tiab] OR "LLM"[tiab] OR "large multimodal  
model"[tiab] OR "LMM"[tiab] OR "multimodal large language  
model"[tiab]) AND (2023:2026[pdat]) AND (english[lang])
```

IEEE Xplore (Institute of Electrical and Electronics Engineers) serves

as the primary repository for the technical *Solution Space*. Given that LMM architectures, Multi-Agent Systems (MAS), and RAG frameworks often debut in conference proceedings before journal publication, IEEE Xplore is essential for capturing the earliest dissemination of algorithmic innovations.

Search String (IEEE Xplore):

```
((("Abstract":"Large language model" OR "Document Title":"Large
language model") OR ("Abstract":"LLM" OR "Document
Title":"LLM") OR ("Abstract":"large multimodal model" OR
"Document Title":"large multimodal model") OR ("Abstract":"LMM"
OR "Document Title":"LMM") OR ("Abstract":"multimodal large
language model" OR "Document Title":"multimodal large language
model")) AND (("Abstract":"clinical decision support system*"
OR "Document Title":"clinical decision support system*"))
```

Web of Science (Clarivate) is included to provide *Integrative Breadth*, functioning as a multidisciplinary anchor that captures the social, ethical, and cross-domain implications of AI deployment. Its comprehensive indexing of citation networks allows for mapping the evolution of Trustworthy AI across diverse scientific fields, bridging the gap between theoretical computer science and practical medical ethics. This selection provides a robust mechanism for identifying multidisciplinary studies that may be absent from specialized indices, thereby minimizing selection bias.

Search String (Web of Science):

```
TS=("Large language model" OR "LLM" OR "large multimodal model"
OR "LMM" OR "multimodal large language model") AND
TS=("clinical decision support system*") AND PY=(2023-2026)
```

3.2.3. Supplementary Searching

To ensure maximal coverage and identify relevant literature that may reside outside the search strategy, a supplementary search will be executed. The supplementary search combines traditional bibliographic techniques with AI-augmented discovery to capture recent technical advancements and regulatory frameworks.

A manual citation snowballing process will be applied to all articles selected for inclusion. Backward citation searching will involve a thorough review of reference lists to identify seminal foundational works. Forward citation searching will be conducted using Google Scholar and Web of Science to identify more recent studies that have cited the primary inclusions.

In recognition of the decentralized nature of current AI research, traditional database queries will be complemented by a semi-automated discovery phase utilizing AI-driven search tools (e.g., Perplexity, Elicit, or Consensus). The specific selection of these tools will be finalized during the execution phase, following iterative testing to determine which platforms yield the highest precision for the technical-clinical intersection.

Every reference identified through AI-augmented methods will undergo strict manual verification by the research team to ensure it meets the established inclusion criteria and aligns with the clinical decision-centric scope. Use of AI discovery tools at this stage is reported per the PRISMA-trAIce commitment in Section 3.1.

3.3. Eligibility Criteria and Study Selection

Studies are screened against the eligibility criteria defined in Section 2 (Population, Concept, Context, and Types of Sources and Evidence). At

the screening stage, additional disqualifying conditions apply: studies are excluded when their technical content is insufficient to characterize the LMM-CDSS architecture, including opinion pieces, abstract-only conceptual works, and studies that reference only legacy transformer architectures (e.g., BERT) without an LMM-CDSS framing.

The selection process follows a two-phase pipeline. Independent human review is the primary mechanism at both phases, with AI-assisted arbitration reserved for records where the two reviewers reach different decisions. In Phase 1, records where either reviewer marks the record as uncertain are also routed to arbitration. The final inclusion or exclusion decision at every stage, along with its documented rationale, is recorded in a structured screening database and processed using Python to ensure full methodological transparency and auditability throughout.

Phase 1: Title and Abstract Screening.

Stage 1: Independent Human Screening. Two reviewers independently assess each record by title and abstract, completing the same structured inclusion/exclusion checklist. Each reviewer records one of three decisions: include, exclude, or uncertain, together with a brief written rationale. Records where both reviewers independently agree to include or exclude are resolved at this stage, with both rationales logged. Records where either reviewer marks the record as uncertain, or where the two reviewers reach different decisions, are forwarded for further assessment.

Stage 2: AI-Assisted Conflict Resolution. For records not resolved in Stage 1, an AI system receives the title and abstract of the record together with the structured checklist responses and rationales from both reviewers.

The AI issues a final include or exclude decision with a documented rationale, which is logged in the screening database. Records the AI classifies as include, along with all records resolved as include in Stage 1, proceed to Phase 2. Use of the arbitration AI is reported per the PRISMA-trAIce commitment in Section 3.1.

Phase 2: Full-Text Screening.

All records that passed Phase 1 are retrieved in full. Records for which the full text cannot be obtained are excluded; these counts and reasons are reported in the PRISMA-ScR flow diagram.

Stage 3: AI-Assisted Human Full-Text Review. Two reviewers independently read each full-text record and apply the same inclusion/exclusion checklist used in Phase 1. At this stage, reviewers must reach a definitive decision: include or exclude. To support a thorough and consistent assessment of complex technical content, reviewers may use AI-assisted document analysis tools (e.g., NotebookLM) as a reading aid to interrogate specific sections of the full text before reaching a decision. The final inclusion or exclusion decision remains the reviewer’s independent judgment. Each decision is accompanied by a documented rationale logged in the screening database. Records excluded at this stage require a specific exclusion reason, which is reported in the PRISMA-ScR flow diagram. Records where the two reviewers reach different decisions are forwarded to Stage 4. Use of reading-aid AI tools is reported per the PRISMA-trAIce commitment in Section 3.1.

Stage 4: AI-Assisted Full-Text Conflict Resolution. For records not resolved in Stage 3, an AI system receives the full text of the paper together with the structured checklist responses and rationales from both

reviewers. The AI issues a final include or exclude decision with a documented rationale logged in the screening database. Use of the arbitration AI is reported per the PRISMA-trAIce commitment in Section 3.1.

3.4. Evidence Charting and Data Management

The extraction process will utilize a standardized table, iteratively refined to capture the intersection of clinical decision archetypes and LMM architectural responses. Data charting will be performed by a single reviewer using the standardized table, with all entries traceable to the source paper to allow subsequent verification by the research team. The final selection of studies will be archived in a version-controlled repository, stored primarily in .csv format to ensure long-term interoperability and immediate accessibility via Python-based analytical frameworks. This dataset functions as the formal evidentiary basis for the review, maintaining a granular audit trail that includes the rationale for inclusion or exclusion at every stage.

The extraction instrument will systematically capture variables across three primary dimensions: (i) *Study Characteristics* (authors, year, clinical specialty); (ii) *The Problem Space* (decision criticality, reasoning trajectory, and workflow integration); and (iii) *The Solution Space* (LMM architecture, use of Multi-Agent Systems, RAG implementation, and XAI functionality).

3.5. Python-Driven Synthesis and Visualization

Following the extraction phase, the raw data will be ingested into a Pandas dataframe for cleaning and synthesis. This computational approach allows for the automated validation of data consistency and the calculation of inter-rater reliability metrics between the two independent human reviewers

at each screening phase, alongside descriptive statistics on the proportion and direction of AI arbitration outcomes.

The final data synthesis will be visualized using Python libraries (e.g., Matplotlib, Seaborn) to produce evidence maps. These visualizations will include a PRISMA-ScR flow diagram detailing the study selection process and multidimensional charts mapping clinical constraints to architectural responses. This programmatic pipeline ensures that the transition from raw search results to the final scoping review is reproducible and aligned with the project’s decision-centric framework.

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